

SAKON NAKHON AGROMET, Thailand

WMO: 483550

Lat: 17.117N

Lon: 104.050E

Elev: 192

StdP: 99.04

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.4	(c) 13.2	(d) 7.9	(e) 6.7	(f) 15.3	(g) 9.4	(h) 7.5	(i) 15.9	(j) 4.2	(k) 24.8	(l) 3.6	(m) 25.3	(n) 0.0	(o) N/A	(p) 0.284

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 10.4	(c) 37.9	(d) 25.8	(e) 36.3	(f) 25.8	(g) 35.0	(h) 26.0	(i) 27.9	(j) 33.1	(k) 27.6	(l) 32.6	(m) 27.3	(n) 32.1	(o) 1.2	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.8	30.3	26.4	22.4	30.1	26.1	22.0	29.9	90.9	32.5	89.3	33.0	87.9	32.5	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.4	3.1	2.5	8.4	39.0	1.7	1.6	7.2	40.1	6.2	41.1	5.2	42.0	4.0	43.2
			7.8	28.9	1.7	0.6	6.6	29.3	5.6	29.7	4.6	30.0	3.4	30.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	22.1	24.4	27.4	29.4	29.1	28.8	28.1	27.9	27.9	26.9	25.3	22.5
	DBStd	3.34	2.89	3.32	3.26	2.76	1.75	1.45	1.70	1.29	1.20	1.41	2.22	2.93
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	13	6	2	1	0	0	0	0	0	0	0	0	4
	CDD10.0	6074	374	403	539	582	591	566	561	554	536	524	458	387
	CDD18.3	3046	122	172	282	332	333	316	303	296	286	266	208	133
	CDH23.3	31324	974	1634	3088	4019	3734	3590	3181	2990	2836	2418	1772	1087
	CDH26.7	12887	313	757	1562	2151	1638	1446	1118	959	946	919	698	381
Wind	WSAvg	0.4	0.5	0.5	0.5	0.5	0.3	0.4	0.5	0.4	0.3	0.3	0.3	0.5
Precipitation	PrecAvg	1615	6	18	48	80	247	299	219	348	263	74	9	4
	PrecMax	2013	47	94	140	127	455	449	364	999	501	268	59	45
	PrecMin	1364	0	0	0	15	136	148	56	107	108	2	0	0
	PrecStd	210	13	27	41	34	87	95	102	209	117	69	16	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.8	36.5	39.1	40.8	37.4	36.0	36.4	34.2	33.9	33.8	34.4	33.6
		MCWB	22.8	23.4	24.5	25.8	27.3	27.1	26.3	27.2	27.3	25.9	25.9	23.9
	2%	DB	31.3	34.7	37.3	38.8	35.8	34.5	34.0	32.9	32.9	32.9	33.0	32.1
		MCWB	22.4	23.4	24.5	26.2	27.1	27.3	27.0	27.0	26.8	25.5	25.1	23.3
	5%	DB	30.0	33.2	35.6	36.8	34.5	33.4	32.6	31.9	32.1	32.0	31.9	30.5
		MCWB	21.8	23.3	24.8	26.4	27.0	27.0	26.7	26.8	26.6	25.4	24.4	22.7
	10%	DB	28.4	31.5	33.9	35.0	33.1	32.4	31.4	30.9	31.2	31.1	30.6	28.8
		MCWB	21.0	22.9	24.8	26.1	26.8	26.8	26.5	26.6	26.3	24.8	23.7	21.7
	0.4%	WB	24.8	25.7	26.7	28.2	28.5	28.3	28.0	28.0	27.7	27.1	26.6	25.0
		MCDB	30.1	32.2	33.3	34.5	34.0	33.1	32.7	32.4	32.1	31.5	32.7	31.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.7	25.0	26.2	27.6	27.8	27.7	27.4	27.4	27.4	26.4	25.8	24.1
		MCDB	28.7	31.8	33.1	34.3	33.2	32.4	31.8	31.4	31.6	30.8	31.1	30.6
	5%	WB	22.7	24.2	25.7	27.1	27.4	27.4	27.0	27.0	27.0	26.0	25.2	23.2
		MCDB	27.6	31.0	32.7	34.2	32.7	31.9	31.2	30.9	31.1	30.3	30.1	29.2
	10%	WB	21.9	23.4	25.2	26.6	27.0	27.0	26.6	26.6	26.6	25.5	24.5	22.3
		MCDB	27.0	29.9	32.1	33.5	32.2	31.4	30.5	30.2	30.3	29.5	28.9	27.8
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.0	11.6	10.4	8.1	6.7	5.9	5.9	6.4	8.4	10.7	12.4
		MCDBR	13.6	14.7	13.1	12.0	9.9	8.3	8.0	7.5	7.9	9.6	11.6	13.2
		MCWBR	5.5	5.2	4.0	3.8	3.5	3.1	3.2	3.1	3.1	3.5	4.4	5.3
	5% WB	MCDBR	11.6	12.6	11.6	10.6	8.7	7.6	6.9	7.0	7.2	8.3	10.2	12.5
		MCWBR	5.2	5.0	4.4	4.0	3.6	3.2	3.2	3.2	3.1	3.3	4.2	5.3
Clear-Sky Solar Irradiance	taub		0.457	0.569	0.660	0.626	0.510	0.468	0.455	0.466	0.504	0.519	0.456	0.453
	taud		2.143	1.853	1.685	1.802	2.148	2.291	2.324	2.281	2.160	2.085	2.236	2.177
	Ebn at Noon		824	747	695	723	801	827	839	837	806	781	820	814
	Edh at Noon		146	205	250	223	156	133	130	136	153	160	133	138
All-Sky Solar Radiation	RadAvg		4.85	5.14	5.15	5.53	5.30	4.91	4.56	4.53	4.64	5.00	4.90	4.66
	RadStd		0.29	0.32	0.22	0.27	0.35	0.36	0.42	0.32	0.36	0.35	0.18	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		+0.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page